

Session objectives

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- Students will be able to describe the basic principles of mechanotransduction found in the inner ear.
- Students will be able to describe the anatomy auditory pathway beginning from the inner ear to the primary auditory cortex.
- Students will be able to describe the basic physiology of the nuclei of the central auditory as well as that of the primary auditory cortex.
- Students will be able to describe the differences between conductive vs. sensorineural hearing loss and provide relevant examples of each as well as how auditory function can be tested by a clinician.

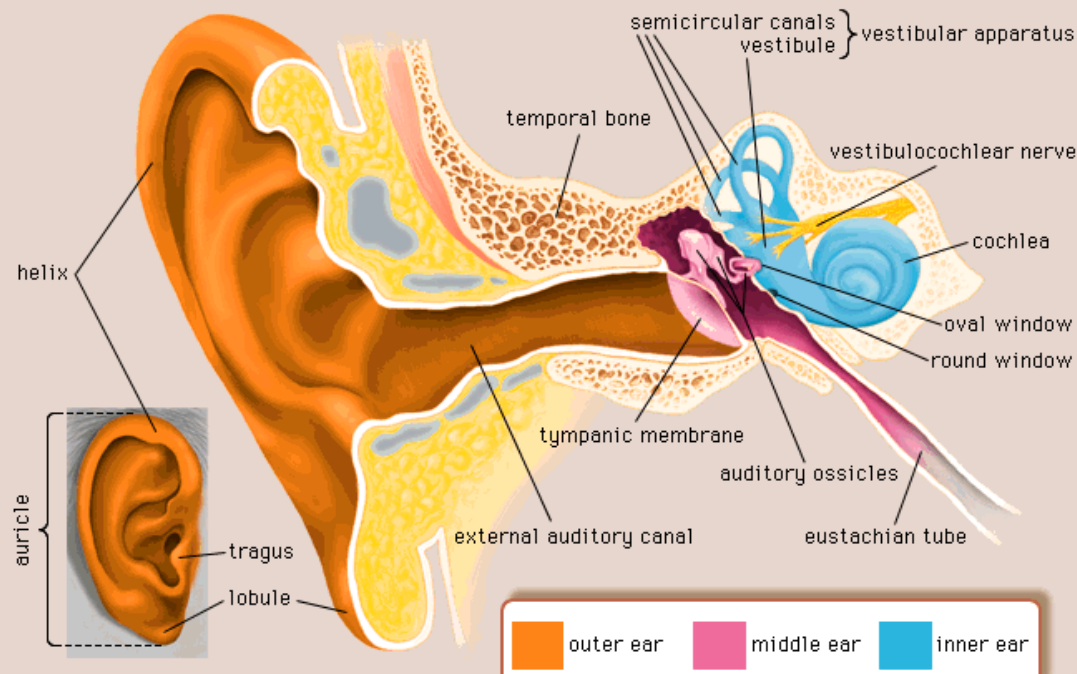


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Auditory anatomy



- Outer ear serves to efficiently capture the sound pressure waves and channel into the middle ear (air transmission).
- In the middle ear, air vibrations of the tympanic membrane are converted into vibrations of the auditory ossicles (solid transmission).
- In the inner ear, the solid vibrations are converted to waves propagating in liquid and converted into nerve impulses.
- Sound pressure of barely audible sound is $20\mu\text{Pa}$ (micro-Pascals) which is 10^8 times weaker than atmospheric pressure.



Middle Ear: Tympanic Membrane & Ossicles

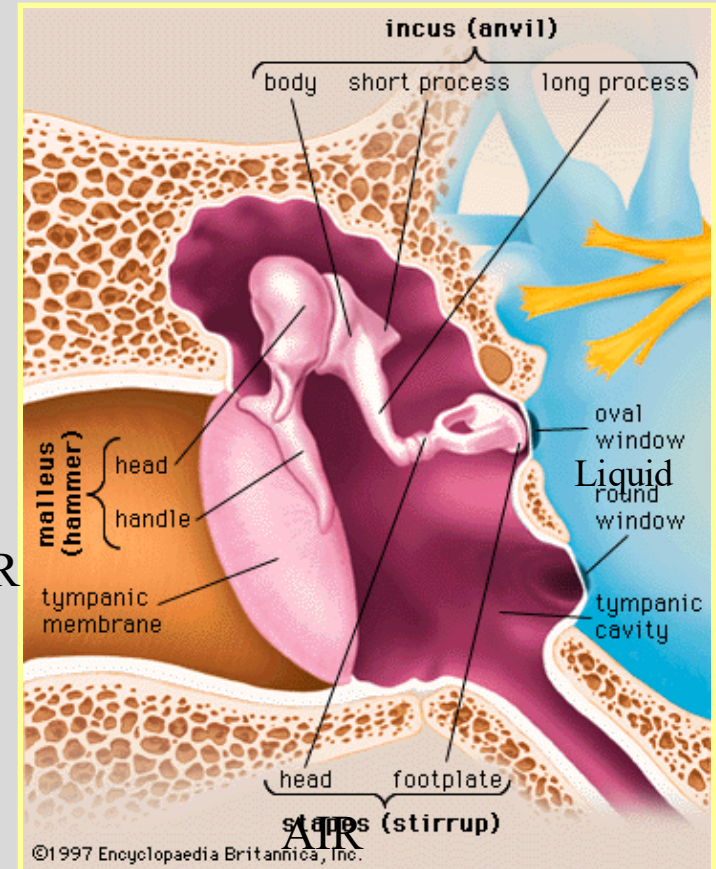
Bones of the middle ear:

Malleus – hammer

Incus – anvil

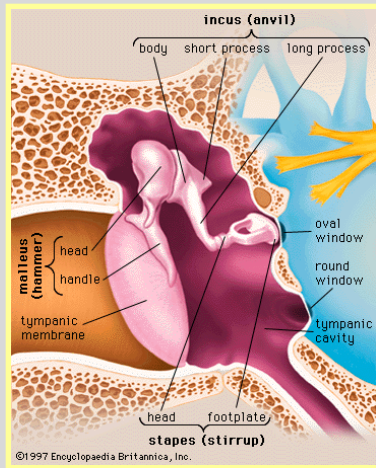
Stapes – stirrup

- Transmit the air vibrations hitting the tympanic membrane to the oval window.
- Smallest bones of the human body.
- Malleus couples to the tympanic membrane.
- Incus transmits the vibration to the stapes.
- The stapes is coupled to the oval window.
- Although the transmission of the vibration in the middle ear is through the ossicles (solid), the tympanic cavity is filled with **AIR** and is connected to the nasopharynx through the eustachian tube.

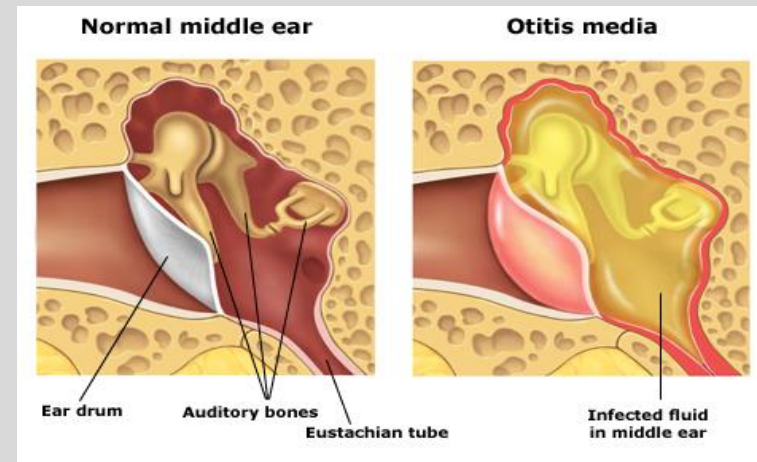


Middle Ear: Conductive hearing losses

- Conditions that affect middle ear conduction
 - Damage to the tympanic membrane
 - Fluid in the tympanic cavity
 - Damage to the ossicles



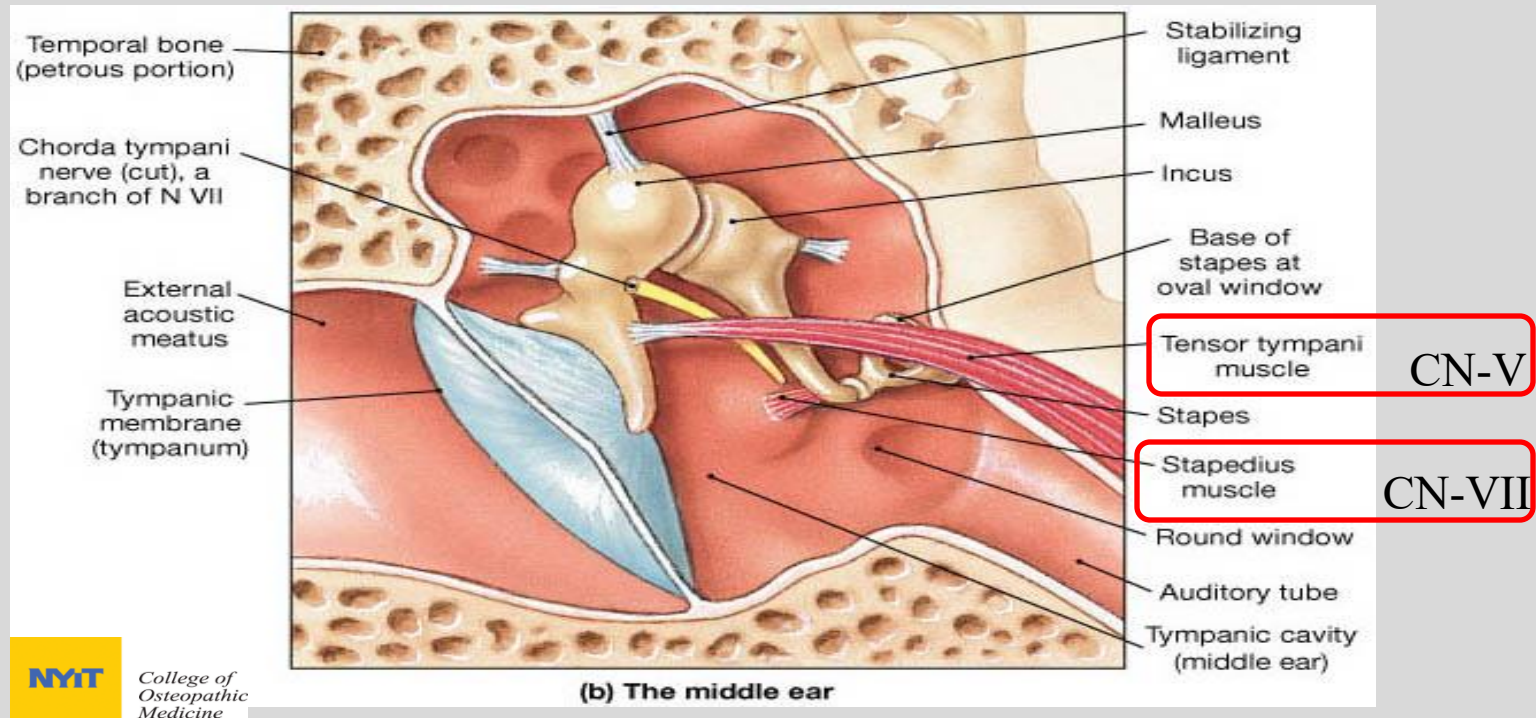
- Perforation of the tympanic membrane equalizes the pressure on both sides, so there is no pressure difference and the tympanic membrane will not vibrate.
- The sound wave applies directly to the oval window without amplification by the ossicles, so there is a significant attenuation of the input to the cochlea.



In otitis media, the tympanic cavity fills with fluid and attenuates the vibrations of the ossicles and the tympanic membrane



Muscles of the middle ear



- Contractions of the muscles of the middle ear attenuates the propagation of sound through the middle ear and protect the delicate hair cells in the inner ear.
- Tensor tympani muscle decouples the malleus from the tympanic membrane and is innervated by the trigeminal nerve.
- Stapedius muscle decouples the stapes from the oval window.
- Both muscles are part of the acoustic reflex to protect the inner ear from loud sounds.
- Sudden loud sounds can still damage the hair cells.

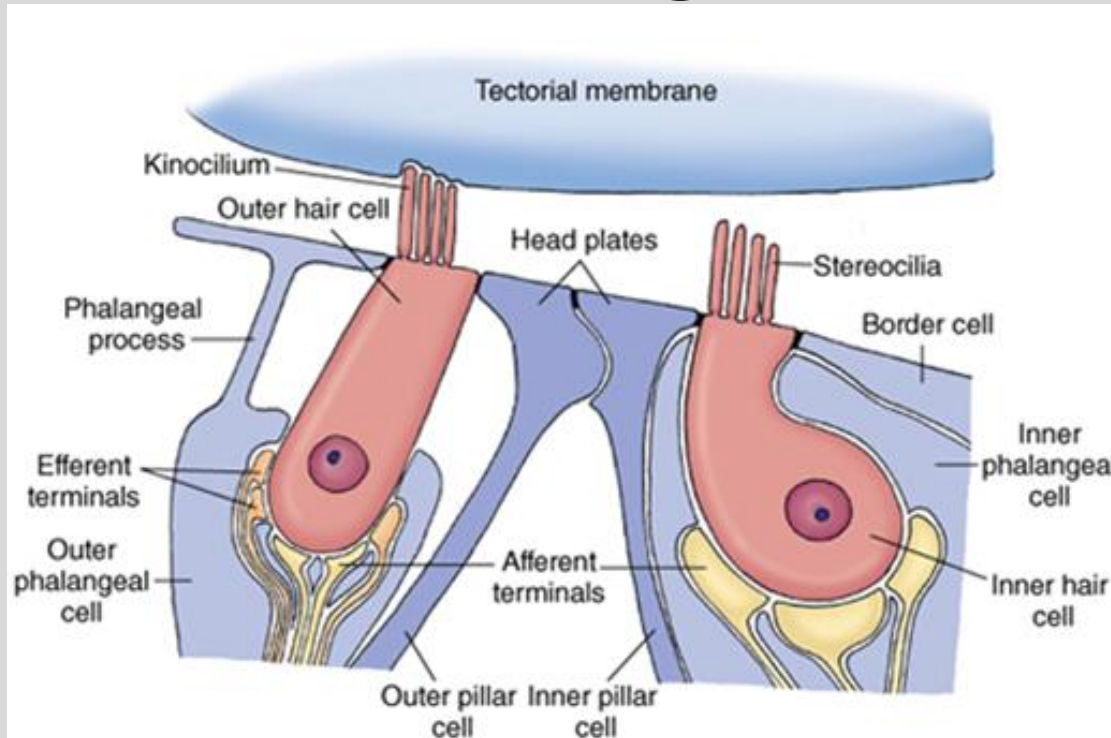


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Organ of Corti

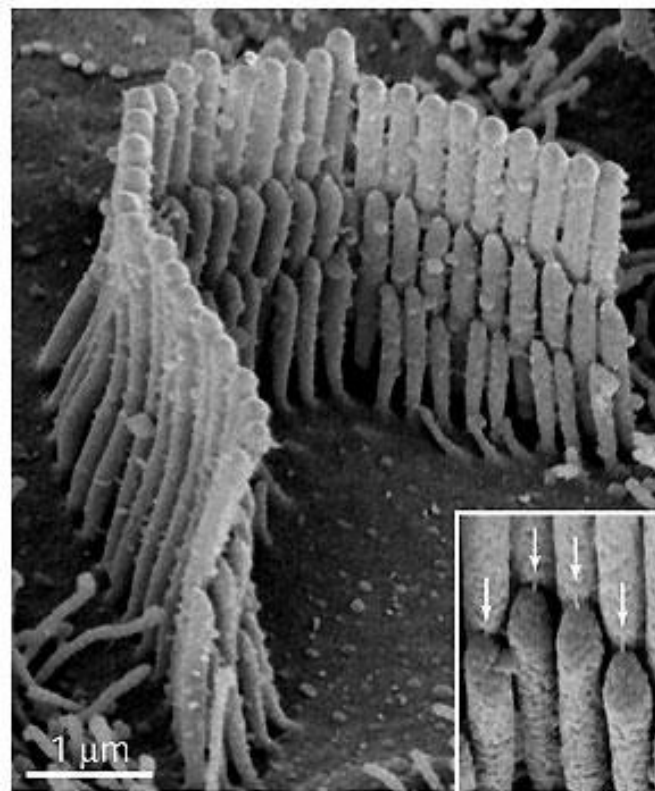
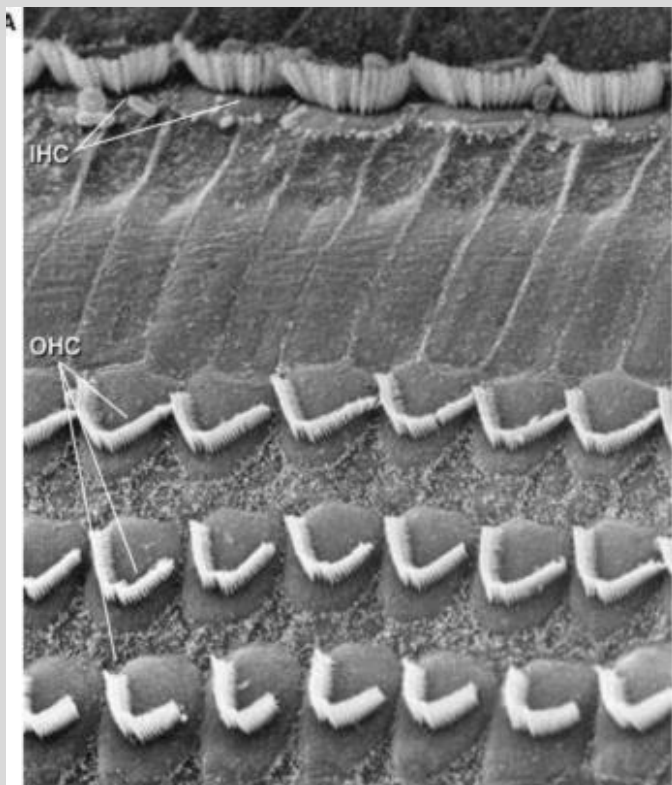


- Two types of hair cells: inner hair cells and outer hair cells.
- The inner hair cells are the actual sensory receptors, and 95% of the fibers of the auditory nerve that project to the brain arise from this subpopulation.
- Stereocilia are sound mechanoelectric transducers found on the tops of hair cells

3000 inner hair cells!!



Inner & Outer hair cells

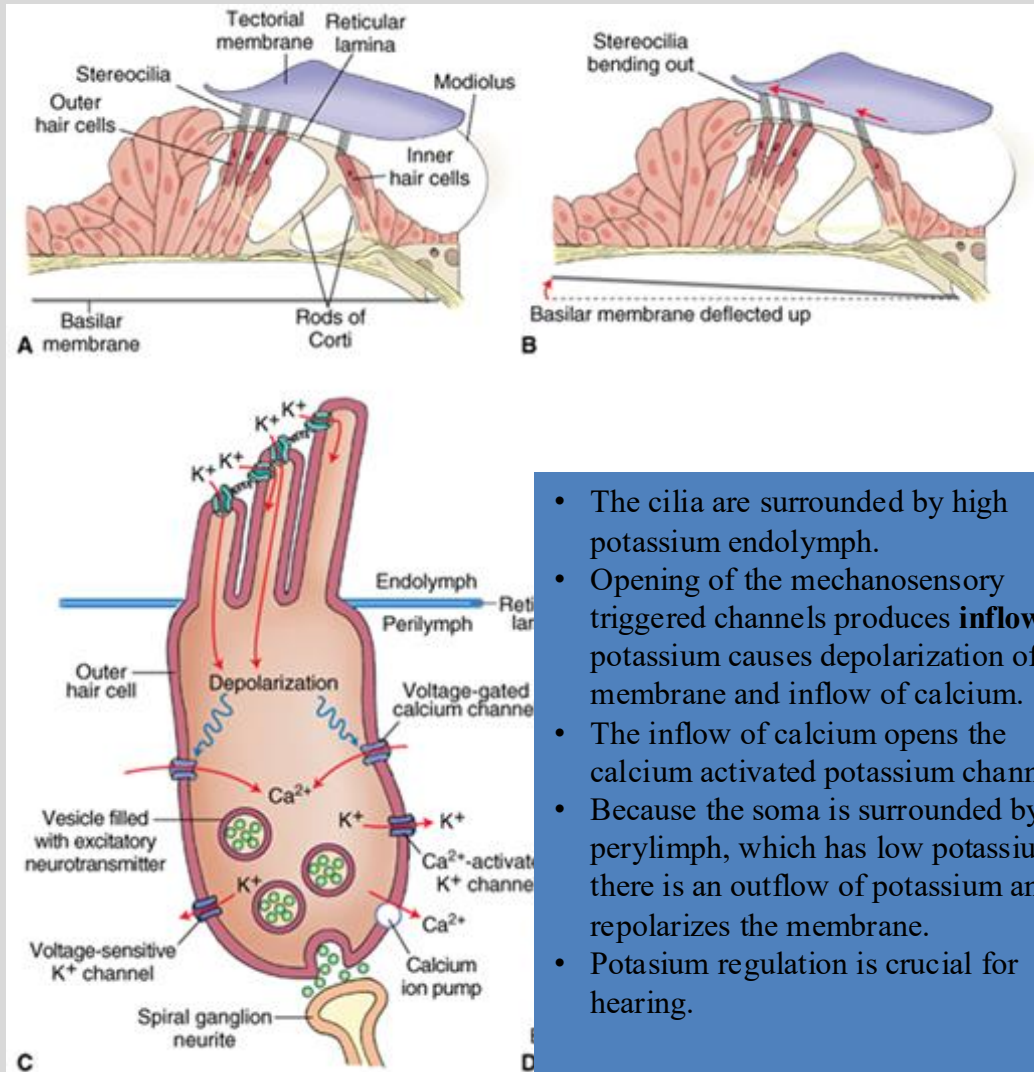


- The cochlear hair cells in humans consist of one row of **inner hair cells** and three rows of **outer hair cells**.
- Very small number of inner hair cells: 3500 inner hair cells.



Hair cells detect deflections of the basilar membrane

- The tips of stereocilia of outer hair cells are embedded in the tectorial membrane.
- Movement of the basilar membrane relative to the tectorial membrane results in displacement of the cilia that is necessary for generating afferent signals.
- Inner hair cells are not attached to the tectorial membrane and movement of their stereocilia is induced by movement of the endolymph.
- Hair cells synapse onto Spiral ganglia neurons
- 1 Spiral ganglia neuron connected to 1 inner hair cell.
- Hair cells release glutamate but do not fire action potentials
- Axons of Spiral ganglia neurons form CNVIII
- Efferent axons also innervate hair cells



- The cilia are surrounded by high potassium endolymph.
- Opening of the mechanosensory triggered channels produces **inflow** of potassium causes depolarization of the membrane and inflow of calcium.
- The inflow of calcium opens the calcium activated potassium channels.
- Because the soma is surrounded by perilymph, which has low potassium, there is an outflow of potassium and repolarizes the membrane.
- Potassium regulation is crucial for hearing.

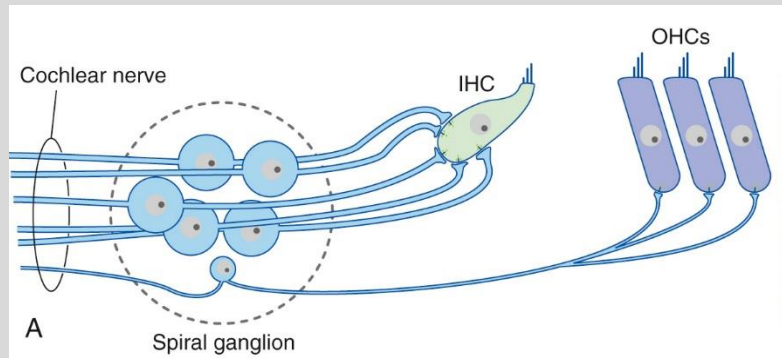


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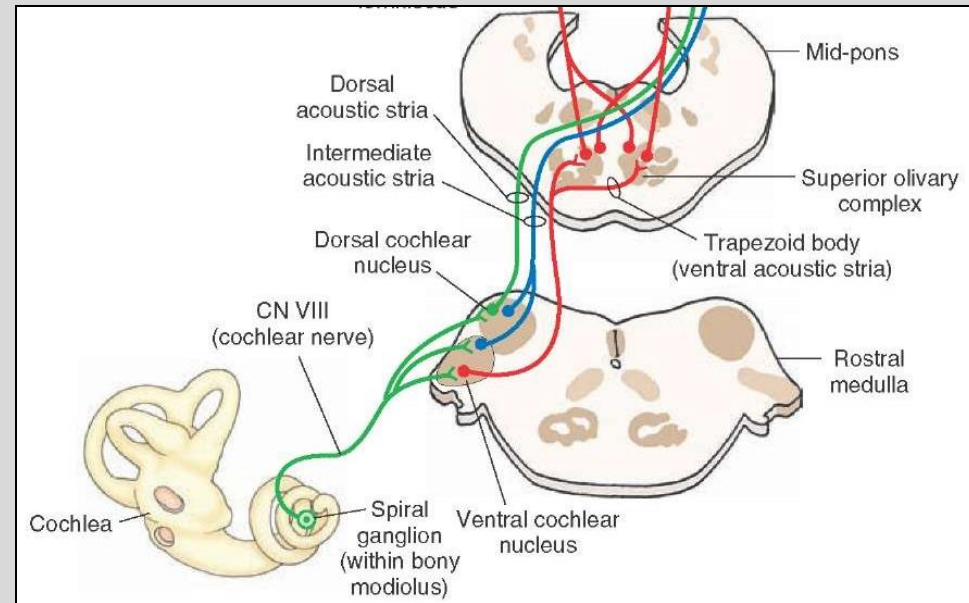
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Ascending auditory pathway



Nolte. The Human Brain

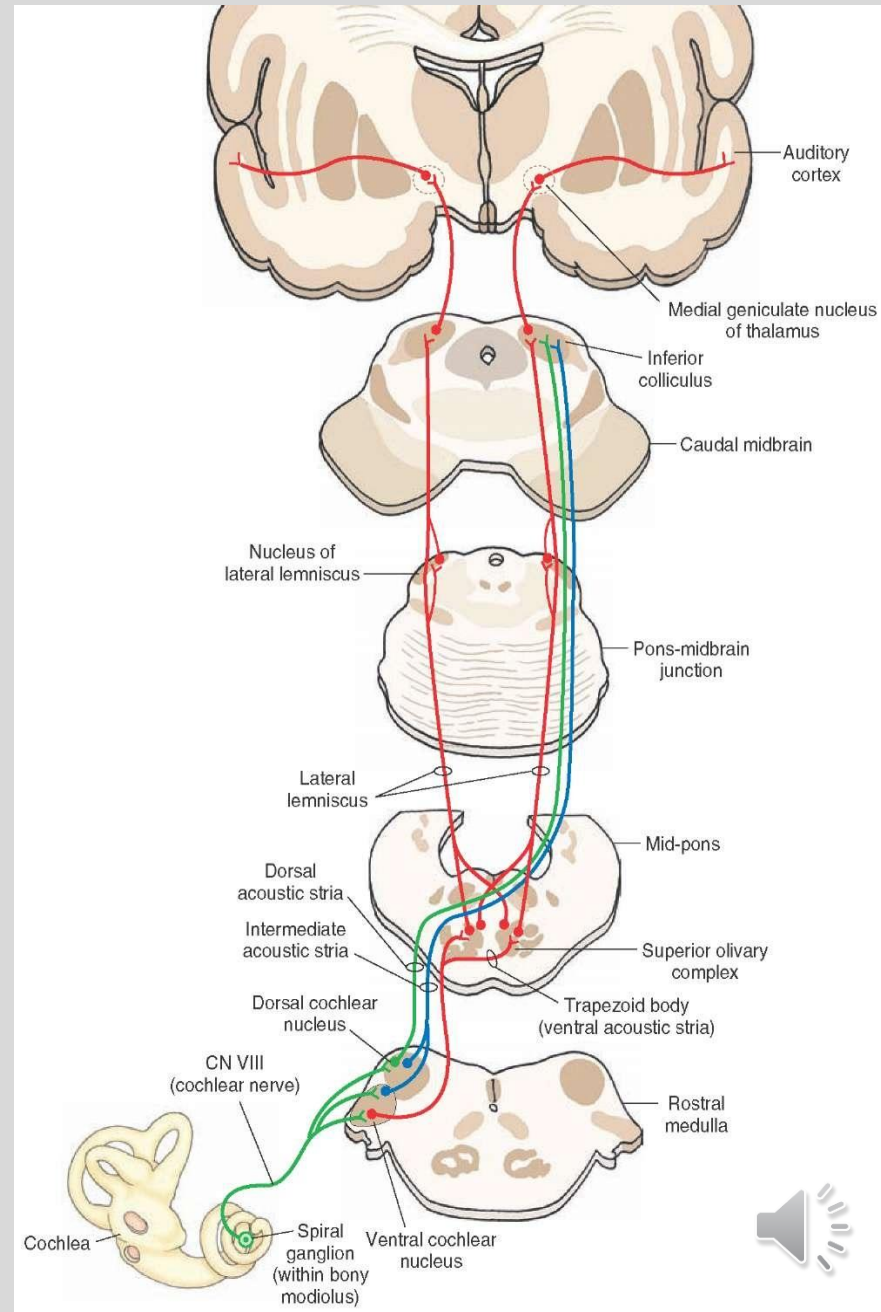


- Neurons in the spiral ganglion receive input from the inner hair cells and send their axons into the ipsilateral cochlear nucleus in the rostral medulla.
- Axons of the cochlear nucleus neurons ascend to the midpons where the fibers decussate and continue ascending.
- Some fibers make synapses in the superior olivary complex in the midpons. Some other fibers decussate and make synapse on the contralateral superior olivary complex. There information from both ears can be compared and sound localization computation can be performed using intensity cues and time differences cues.



Auditory pathway

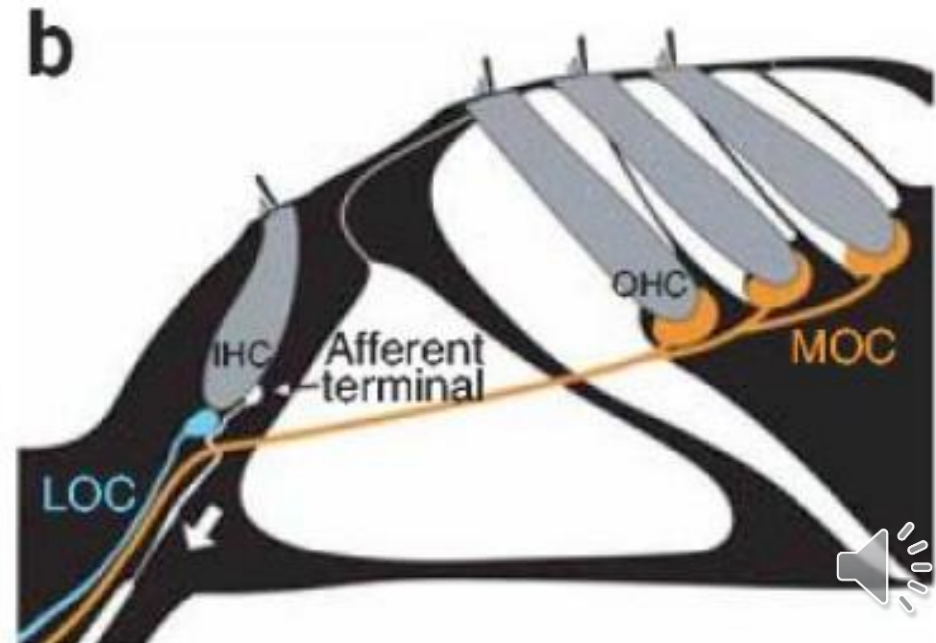
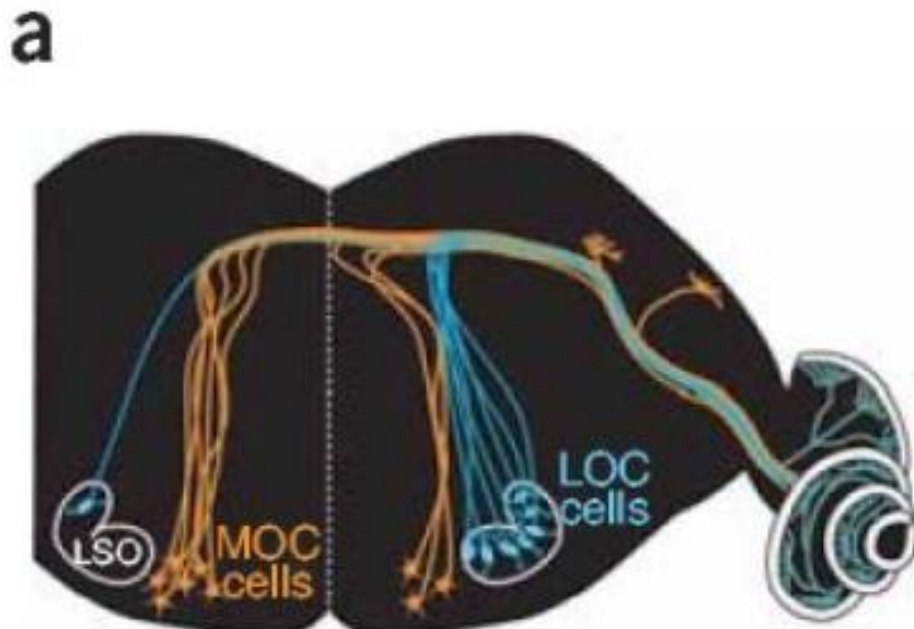
- Fibers ascend through the lateral lemniscus.
- After the decussation in the midpons, the fibers carrying auditory information make synapses in the inferior colliculus.
- The inferior colliculus project to the medial geniculate body of the thalamus bilaterally (auditory thalamus)
- Thalamic neurons project to the auditory cortex in the temporal lobe.



Olivocochlear descending pathway - two major subsystems

The medial (MOC) component is made up of myelinated axons of neurons of the medial superior olive projecting to the cochlear outer hair cells.

The lateral (LOC) component is made up of unmyelinated axons of neurons of the lateral superior olive projecting to cochlear nerve fibers, near their afferent synapses with cochlear inner hair cells.



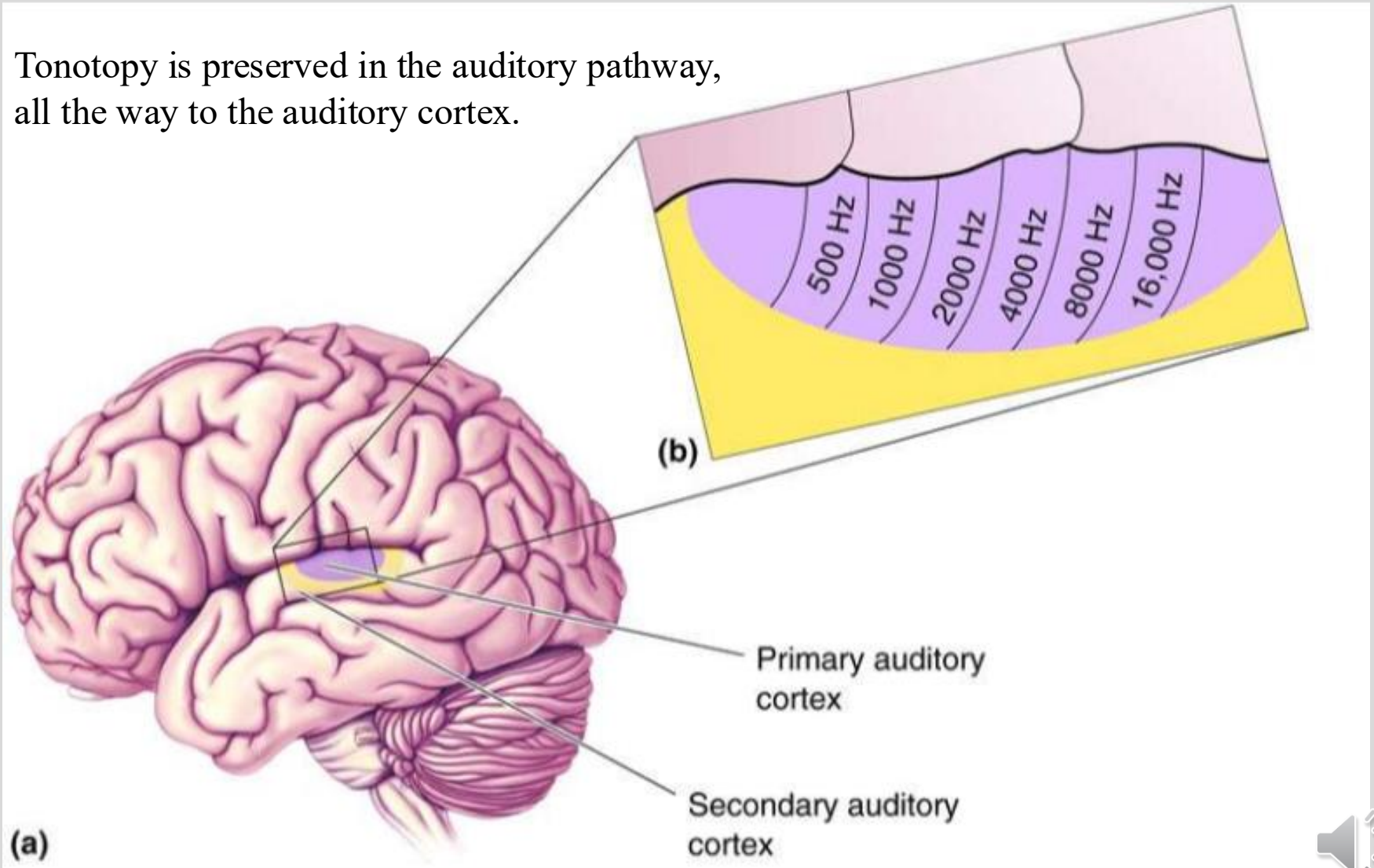
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Tonotopy is preserved in Auditory Cortex

Tonotopy is preserved in the auditory pathway,
all the way to the auditory cortex.



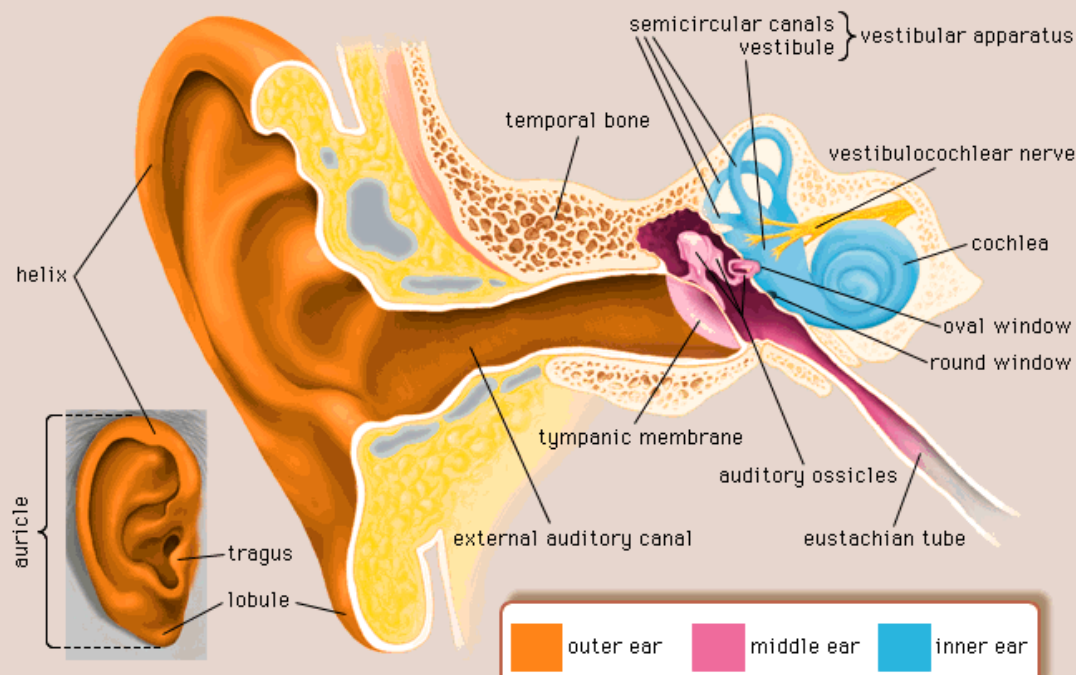
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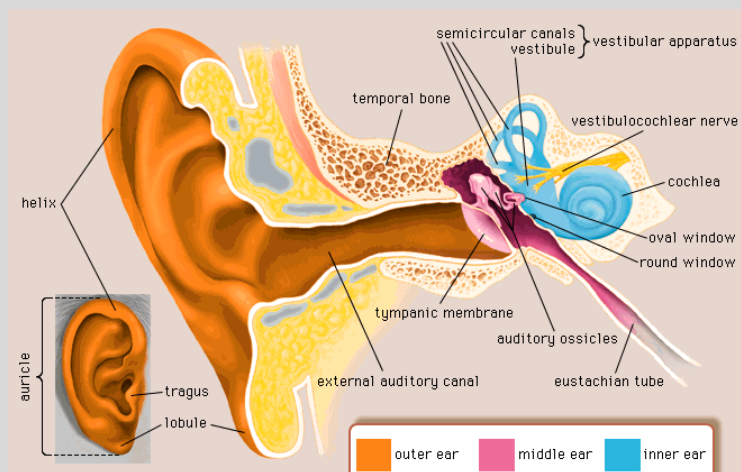
Hearing Loss

- **Conductive:** involves anything that blocks the conduction of sound from the external and middle ear
 - This is often correctable
- **Sensorineural:** involves the inner ear, cochlea, or auditory nerve
 - Usually caused by hair cell pathology
- **Mixed Loss:** Combination of the two



Causes of Conductive Hearing Loss

- Congenital – atresia or malformation of the external auditory canal



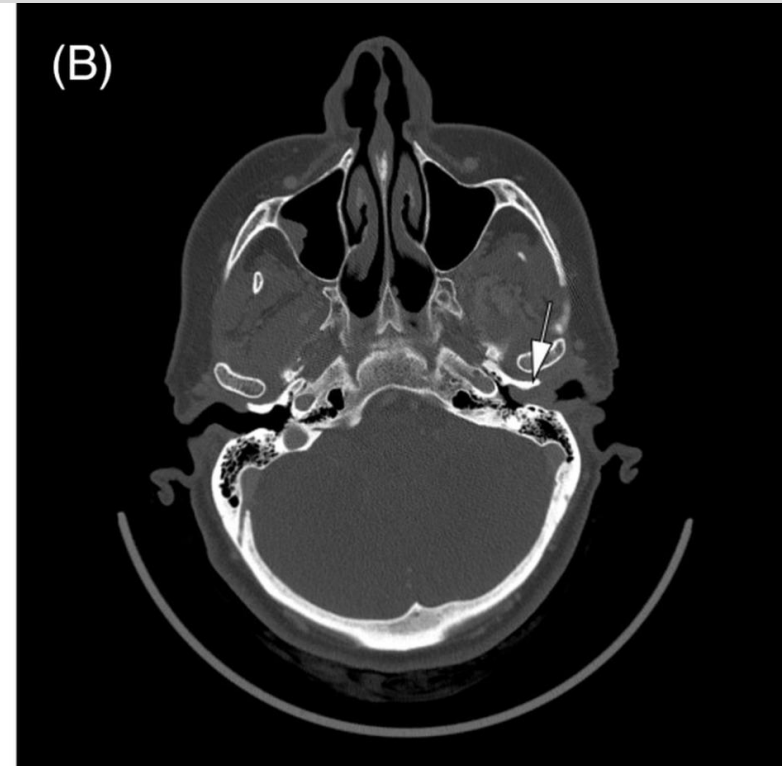
. Atresia without microtia. (Courtesy of Glenn Isaacson, MD, Philadelphia, PA.)

- It is unilateral in a majority of cases
- Children with unilateral aural atresia typically have normal speech development so long as the other ear is unaffected and typically have normal hearing and ear development in the unaffected ear.
- Possibility of malformation of the ossicles
- Treatment is surgical
- They are at increased risk of delayed language development due to functional mono-aural hearing, and early identification of this is essential.



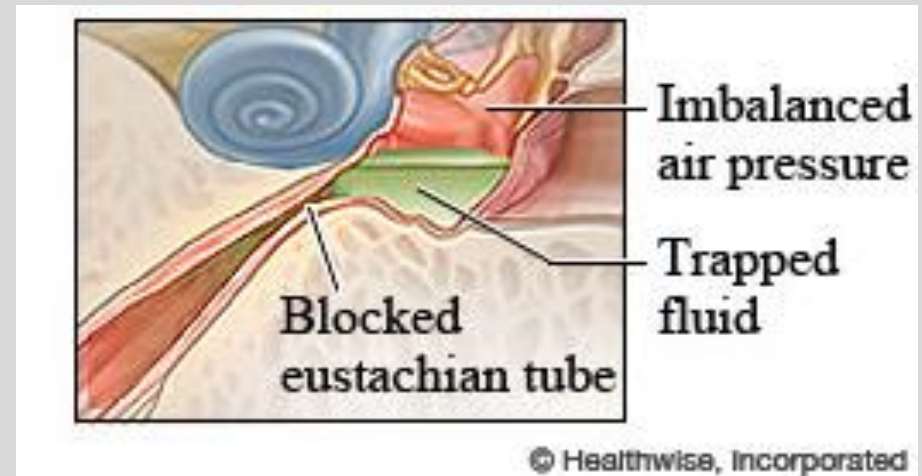
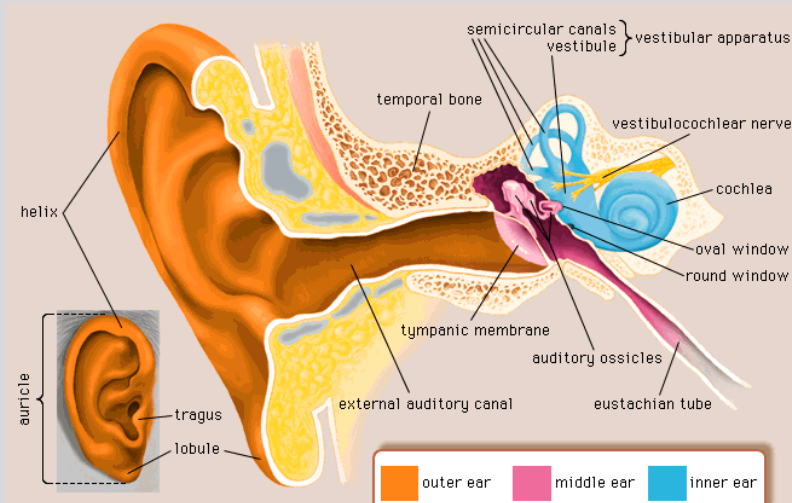
Causes of Conductive Hearing Loss

- Squamous cell carcinoma can block the ear canal or affect the middle ear



Causes of Conductive Hearing Loss

- Eustachian tube dysfunction – usually with viral URI or sinusitis

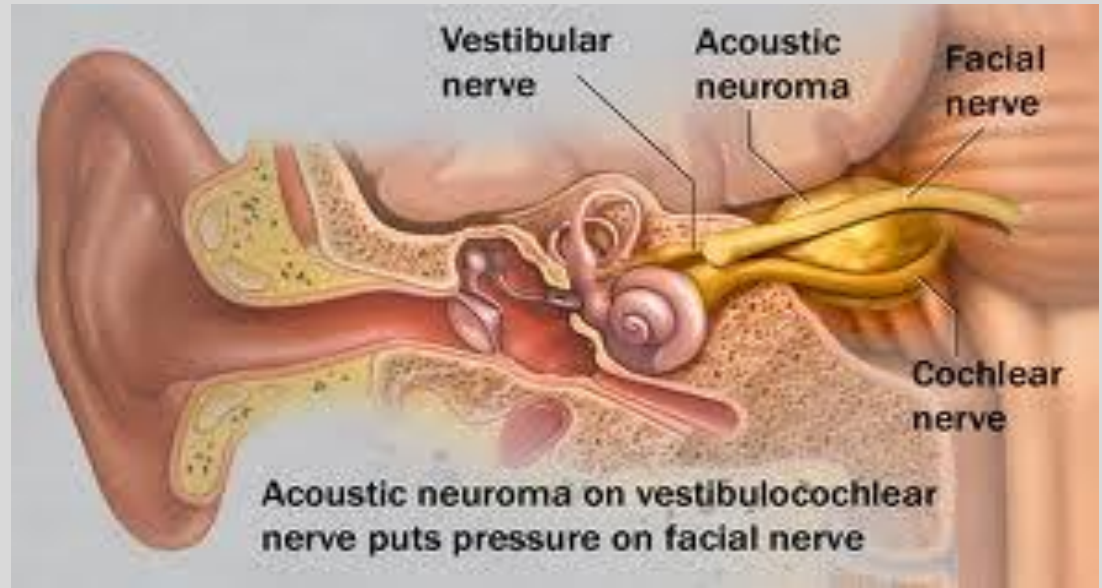


- Blocked eustachian tube leads to the accumulation of fluid in the middle ear.
- The fluid accumulation affects the transmission of vibration thorough the ossicles.
- Middle ear barotrauma – filling of middle ear with serous fluid or blood



Vestibular schwannoma

4/100,000



- Slow growing tumor of the vestibular cochlear nerve
- Benign tumor, but can cause damage to surrounding structures as it grows including facial nerve
- Most common symptoms: vertigo, hearing loss, tinnitus
- Most useful test to diagnose: MRI of the brain
- Surgery is the treatment
 - Complications include hearing loss and paralysis of the facial muscles



Ear & Auditory testing

- Otoscopy exam
- Pure-Tone Testing
- Speech Testing
- Tests of the Middle Ear
 - tympanometry, acoustic reflex testing
- Auditory Brainstem Response
- Otoacoustic Emissions
- Weber & Rinne tests



Summary slide

- The anatomy of the auditory system includes the ear (outer, middle, inner) as well as diverse central nuclei of the brainstem and thalamus and finally the neocortex.
- Audition emerges from mechanotransduction and synaptic transmission along the auditory pathway.
- Dysfunction of mechanotransduction or synaptic transmission broadly causes hearing loss.

